Employee Leave Management (ELM)

Comprehensive Technical Specification & Phase-2 Engineering Master Reference

SYSTEM / PROJECT	PHASE FOCUS	SPECIFICATION BASELINE	DEPLOYMENT TARGET
Fusion ERP (Assignment 6B)	Phase 2: Review, Test & Improve	Final Synchronized Baseline v1.4 / v1.6	Integrated with Fusion Base Code

Document Overview & Architectural Baseline

This technical specification provides comprehensive engineering documentation for the Employee Leave Management (ELM) module in Fusion ERP. It consolidates all authoritative requirements across **Authentication & Authorization**, **REST API Integration**, **Relational Database Integration**, **Business Rules (BR-EL-001..038)**, **Use Cases (EL-UC-001..021)**, **Workflows (BW-EL-01..17 / CW-EL-01..04)**, and the **Phase-2 Automated Unit Testing Strategy**.

1. Authentication & Authorization Integration

1.1 Centralized Stateless Authentication (Fusion IAM)

Individual ERP modules in Fusion do not maintain user password hashes, manage credentials, or store session states. Authentication is completely decoupled and handled by the central identity service (`Fusion_System_Administrator` / IAM).

Incoming client requests are authenticated via `IamSessionAuthentication` (DRF authentication handler), supporting two transport mechanisms:

- **Browser / SPA Ambient Cookie:** Read from the `httpOnly` cookie defined in `settings.IAM_AUTH_COOKIE_NAME`. Requires mandatory CSRF token verification.
- **API / Script Bearer Token:** Extracted from the `Authorization: Token <token>` HTTP header. CSRF is bypassed for token headers.

Both mechanisms resolve the token synchronously via `get_client().resolve_session(auth_token=token)`. Upon resolution, the request receives an immutable `Principal` object on `request.principal`.

1.2 The Principal Object & Module Boundary Isolation

The `Principal` value object encapsulates identity and authorization attributes without coupling the database to Django's default User model:

```
@dataclass(frozen=True)
class Principal:
    user_id: int                 # Plain integer index stored in module tables
    username: str                # User login identity
    display_name: str            # Full name for audit and UI display
    kind: str                    # faculty | staff | student
    active_role: str | None      # Currently active role
    roles: tuple[str, ...]       # All assigned roles
    permissions: frozenset[str]  # Resolved permissions set
    modules: tuple[str, ...]     # Authorized module list
    navigation: tuple[dict, ...] # Dynamic sidebar items
    email: str =
```

Architectural Invariant: Zero Cross-Module Foreign Keys

Per Fusion ERP Architectural Rule (enforced by `ops/checks/no_cross_module_fk.py`), the ELM module is strictly forbidden from referencing User or Profile models via `models.ForeignKey`. All user associations (applicant, substitute, Unit Head, sanctioning authority) are strictly persisted as plain integer indices (`user_id = models.IntegerField(db_index=True)`).

1.3 Permissions Catalog & Role-Based Access Control (RBAC)

The ELM module registers fine-grained permissions in `modules/leave/registry.py` and enforces them using DRF permission classes `HasModuleGrant(leave_module)` and `HasPermission("leave_module.*")`:

Permission Code	Description & Operational Scope	Default Role Grants
`leave_module.request.create`	Apply for new leaves, submit resumption, request cancellation or extension, and withdraw pending requests.	FACULTY STAFF STUDENT HOD DEAN REGISTRAR DIRECTOR
`leave_module.request.view_self`	Inspect personal leave balances, historical applications, status timeline, and export printable applications.	FACULTY STAFF STUDENT ALL APPROVERS LEAVE ADMIN
`leave_module.substitute.respond`	Access substitute inbox; accept or decline nominated responsibility assignments.	FACULTY STAFF STUDENT ALL APPROVERS
`leave_module.request.review`	Unit Head recommendation/sanction, higher-authority final sanction decisions, and resumption review.	HOD DEAN REGISTRAR DIRECTOR

Permission Code	Description & Operational Scope	Default Role Grants
`leave_module.admin.manage`	Maintain policy parameters, manage holiday/vacation calendars, sync offline leaves, and run year-end processing.	LEAVE ADMINISTRATOR ACADADMIN

2. API Integration

2.1 REST API Surface (Backend Endpoints)

All endpoints are mounted under `/api/v1/leave/` and strictly return structured JSON payloads:

Path	Method	Required Permission	Payload / Query Parameters	Status Codes
`/leave/apply/`	POST	`request.create`	`{ category, start_date, end_date, reason, half_day?, substitute_user_id?, station_leave?, station_address?, medical_certificate_url? }`	201 Created 400 Bad Request
`/leave/mine/`	GET	`request.view_self`	Query: `?status=<LeaveStatus>` (optional)	200 OK
`/leave/balances/`	GET	`request.view_self`	Query: `?year=<YYYY>` (defaults to current year)	200 OK
`/leave/<id>/`	GET	`request.view_self`	None (checks ownership or approver assignment)	200 OK / 403 / 404
`/leave/<id>/withdraw/`	POST	`request.create`	`{ reason: string }`	200 OK / 400 Bad
`/leave/<id>/cancel/`	POST	`request.create`	`{ cancellation_reason: string }`	200 OK / 400 Bad
`/leave/<id>/extend/`	POST	`request.create`	`{ extended_end_date: string, reason: string }`	200 OK / 400 Bad
`/leave/<id>/resume/`	POST	`request.create`	`{ resumption_date: string, remarks?: string }`	200 OK / 400 Bad
`/leave/<id>/print/`	GET	`request.view_self`	None (returns formatted export document)	200 OK
`/leave/substitute/`	GET	`substitute.respond`	Filters requests where `substitute_user_id == current_user`	200 OK
`/leave/substitute/<id>/respond/`	POST	`substitute.respond`	`{ accept: boolean, remarks?: string }`	200 OK / 400 Bad
`/leave/substitute/<id>/replace/`	POST	`request.create`	`{ substitute_user_id: number }`	200 OK / 400 Bad
`/leave/approvals/`	GET	`request.review`	Query: `?status=<LeaveStatus>` (approver queue)	200 OK
`/leave/approvals/<id>/unit-head/`	POST	`request.review`	`{ decision: "APPROVE"|"REJECT"|"RECOMMENDED"|"NOT_RECOMMENDED", remarks?: string }`	200 OK / 400 Bad
`/leave/approvals/<id>/sanction/`	POST	`request.review`	`{ approved: boolean, remarks?: string }`	200 OK / 400 Bad
`/leave/approvals/<id>/resumption/`	POST	`request.review`	`{ remarks?: string }`	200 OK (adjusts ledger)
`/leave/calendar/`	GET	Authenticated	Query: `?year=<YYYY>`	200 OK
`/leave/vacation-periods/`	GET	Authenticated	Query: `?year=<YYYY>`	200 OK
`/leave/admin/offline-sync/`	POST	`admin.manage`	`{ user_id, category, start_date, end_date, reason, offline_reference }`	201 Created
`/leave/admin/year-end/`	POST	`admin.manage`	`{ year: number }`	200 OK (batch log)

2.2 Cross-Module Plural Contracts (Inter-Module Gateway)

To preserve modular isolation, external modules (Payroll, Attendance, Administration) access leave data only via batch-oriented plural functions defined in `modules/leave/contracts.py`:

```
def get_leave_requests(request_ids: Sequence[int]) -> list[dict]:
    """Retrieve multiple leave requests by ID in a single query without exposing models."""

def get_user_leave_balances(user_ids: Sequence[int], year: int) -> dict[int, list[dict]]:
    """Batch-query annual leave ledgers for payroll deduction or attendance synchronization."""
```

3. Database Integration & Data Model

3.1 Data Architecture & Schema Definition

The ELM persistence layer is designed for relational consistency, auditable state transitions, and zero cross-module coupling. The database tables are defined as follows:

Table & Model	Key Columns & Types	Constraints, Indexes & Business Purpose

leave_balances
`LeaveBalance`
	`user_id: int`
	`year: smallint`
	`category: varchar(10)`
	`credited: decimal(5,1)`
	`availed: decimal(5,1)`
	`balance: decimal(5,1)`
		• `unique_together = ('user_id', 'year', 'category')`
		• Index on `(user_id, year)`
		• Maintains running annual ledger per employee and category.

leave_requests
`LeaveRequest`
	`user_id: int`
	`category: varchar(10)`
	`start_date, end_date: date`
	`half_day: varchar(15)`
	`computed_days: decimal(5,1)`
	`status: varchar(30)`
	`substitute_user_id: int`
	`substitute_status: varchar(20)`
	`station_leave: boolean`
	`unit_head_recommendation: str`
	`offline_entry: boolean`
		• Indexes: `(user_id, status)`, `(substitute_user_id, substitute_status)`, `(start_date, end_date)`
		• Core lifecycle entity representing applications, coupled station leave, approvals, extensions, and resumptions.

leave_action_logs
`LeaveActionLog`
	`leave_request_id: int`
	`actor_user_id: int`
	`action: varchar(40)`
	`from_status: varchar(30)`
	`to_status: varchar(30)`
	`remarks: text`
	`created_at: timestamp`
		• Index on `leave_request_id`
		• Immutable append-only audit trail recording every state change, actor endorsement, and decision timestamp.

leave_holiday_calendar
`HolidayCalendar`
	`date: date (UNIQUE)`
	`name: varchar(100)`
	`is_restricted: boolean`
	`year: smallint`
	`published: boolean`
		• Authoritative calendar source for closed holidays and Restricted Holidays (RH). Date uniqueness enforced.

leave_vacation_periods
`VacationPeriod`
	`name: varchar(100)`
	`start_date, end_date: date`
	`year: smallint`
		• Academic Calendar vacation periods. Used to validate faculty Vacation Leave (VL) requests (BR-EL-006).

leave_policy_configs
`LeavePolicyConfig`
	`category: varchar(10) (UNIQUE)`
	`annual_entitlement: decimal`
	`carries_forward: boolean`
	`max_carry_forward: decimal`
	`continuous_counting: boolean`
		• Dynamic policy parameters governing annual entitlement, carry-forward rules, and day counting methods.

leave_sla_configs
`SLAConfig`
	`task_type: varchar(40) (UNIQUE)`
	`reminder_hours: int`
	`escalation_hours: int`
	`escalation_target_role: varchar`
		• Configuration for SLA-driven automated reminders (BR-EL-032) and supervisory escalation (BR-EL-033).

3.2 Concurrency & Transaction Management

To prevent race conditions during concurrent leave submissions or approvals:

- All balance debits, status transitions, and audit logging execute inside `django.db.transaction.atomic()` blocks.
- Leave ledger queries use `select_for_update()` to lock the specific employee balance row during balance verification and deduction.

4. Complete Business Rules (BR-EL-001 through BR-EL-038)

Rule ID	Name & Category	Authoritative Rule Statement & Operational Outcome
BR-EL-001	Category Eligibility ELIGIBILITY	A new leave request shall use only CL, RH, SCL, EL, COL, or VL and satisfy category eligibility (e.g. VL is strictly faculty-only). Reject invalid categories upfront.
BR-EL-002	CL Entitlement and Lapse ENTITLEMENT	Each employee receives 8 CL days per calendar year. Unused CL completely lapses at year-end (December 31); strictly no carry-forward.
BR-EL-003	RH Entitlement & Calendar CALENDAR	Each employee receives 2 RH days per year. The requested date must match an applicable holiday marked `is_restricted=True` in the published calendar. Unused RH lapses at year end.
BR-EL-004	SCL Entitlement & Scope AUTHORITY	Up to 15 SCL days per year for admissible duty. Purpose admissibility is judged by the Sanctioning Authority (no hardcoded admissibility table). Unused SCL lapses.

Rule ID	Name & Category	Authoritative Rule Statement & Operational Outcome
BR-EL-005	Non-Faculty EL Entitlement ENTITLEMENT	Non-faculty employees receive 30 EL days per year. Unused balance carries forward across year boundaries according to configured limits.
BR-EL-006	Faculty Vacation Leave (VL) CALENDAR	Faculty members receive 60 VL days/year. VL may be availed **only** during vacation periods specified in the Academic Calendar.
BR-EL-007	VL-to-EL Year-End Conversion YEAR-END	At year end, all unused faculty VL converts to EL at a **2 VL : 1 EL** ratio (including fractional conversions). VL itself never carries forward as VL.
BR-EL-008	Treatment of Converted EL ACCOUNT	EL created through VL conversion is credited directly to the faculty member's EL balance ledger, carries forward, and can subsequently be availed as normal EL.
BR-EL-009	Commuted Leave (COL) EVIDENCE	Eligible employees have 20 COL days/year. COL is medical/balance-controlled; supporting medical certificate URL is mandatory. Unused COL carries forward.
BR-EL-010	Half-Day CL Treatment CALCULATION	Half-day leave is supported **only** for Casual Leave (CL). Requires `start_date == end_date`. Consumes exactly **0.5 days** from the CL balance.
BR-EL-011	Continuous-Day Counting COUNTING	• For **EL, COL, and VL**: Intervening weekends (Sat/Sun) and closed holidays inside the approved interval COUNT as leave days. • For **CL, SCL, and RH**: Intervening weekends and closed holidays do NOT count as leave days.
BR-EL-012	Applicant Non-Overlap VALIDATION	A new request cannot overlap active (pending or approved) leave. **Exception:** Complementary half-day CL requests on the same single date (one First Half, one Second Half) are non-overlapping.
BR-EL-013	Station Leave Coupling COUPLING	Station Leave permission is captured as embedded fields within the primary leave application (station address, emergency contact, station dates). It is not a separate application.
BR-EL-014	Substitute Eligibility SUBSTITUTE	When coverage is required: exactly one substitute; substitute must be an active employee; cannot be the applicant; must have no overlapping pending or approved leave.
BR-EL-015	Substitute Consent Gate WORKFLOW GATE	Approval routing cannot proceed until nominated substitute accepts. If declined, gate stays closed; applicant may nominate an eligible replacement or withdraw.
BR-EL-016	Unit Head Final Scope (CL/RH) AUTHORITY	For CL and RH (and coupled Station Leave), the applicant's Unit Head (HoD) makes the **final** Approve or Reject decision (except Director self-sanction).
BR-EL-017	Unit Head Recommendation AUTHORITY	For SCL, EL, COL, and VL, Unit Head records `RECOMMENDED` or `NOT_RECOMMENDED`. Neither is a final rejection; both forward to higher sanction.
BR-EL-018	Higher-Sanction Routing WORKFLOW	Higher-sanction requests route from Unit Head through configured hierarchy (Establishment / Dean / Registrar) to the resolved competent Sanctioning Authority.
BR-EL-019	Authority Resolution HIERARCHY	Routing resolved from applicant designation, department, and category. Establishment step applies only where explicitly configured.
BR-EL-020	Director Self-Sanction EXECUTIVE	When the Director is the applicant, Unit Head and higher authority steps are bypassed; request auto-sanctions (subject to substitute acceptance if required).
BR-EL-021	Retired Identifier RETIRED	Former rule permitting in-place modification of pending requests is retired. Superseded by BR-EL-034.
BR-EL-022	Immediate Withdrawal LIFECYCLE	A pending request may be withdrawn immediately at any point before final approval. Halts remaining processing, blocks further decisions, and frees dates/balance.
BR-EL-023	Approved Leave Cancellation CANCELLATION	Approved leave can be cancelled **only before** its start date. Routes through original hierarchy. Approval marks request `CANCELLED` and refunds balance.
BR-EL-024	Fresh Application After Cancel LIFECYCLE	If revised leave is needed after cancellation, a fresh application must be submitted rather than modifying the cancelled request.
BR-EL-025	Ongoing Leave Extension EXTENSION	Permitted only for ongoing EL, COL, or VL (`start_date <= today <= end_date`). Approval posts only the incremental additional period to balance and records.
BR-EL-026	Retired Identifier RETIRED	No active rule assigned to this identifier in final baseline.
BR-EL-027	Resumption Applicability RESUMPTION	Formal return-to-duty review applies to EL, COL, and VL. Employee submits actual resumption date; Sanctioning Authority reviews.
BR-EL-028	Resumption Adjustments LEDGER SYNC	• Normal Return (`resumption == end_date`): No balance adjustment. • Early Return (`resumption < end_date`): Unused days refunded to balance ledger. • Late Return (`resumption > end_date`): Excess days deducted from balance ledger.
BR-EL-029	Offline Leave Synchronization ADMIN SYNC	Retrospectively records leave already sanctioned on paper. Validates balance, deducts ledger, marks directly `APPROVED`; triggers no approval workflow.
BR-EL-030	Configured Policy Hierarchy CONFIG	Validation and routing use currently active policy parameters. No dynamic publishing lifecycle overhead in this baseline.
BR-EL-031	Effective Published Calendar CALENDAR	Date and RH validations strictly read from the published holiday calendar until an updated calendar is published.

Rule ID	Name & Category	Authoritative Rule Statement & Operational Outcome
BR-EL-032	SLA Reminder on Pending Task SLA	Issues reminder notification to assigned user when an actionable task remains pending past configured reminder threshold hours.
BR-EL-033	SLA Escalation on Pending Task SLA	When an actionable task passes escalation threshold hours, notifies higher authority and re-routes task responsibility if configured.
BR-EL-034	No Post-Submission Modification LIFECYCLE	A submitted pending leave request cannot be edited in place. Any change requires withdrawal and resubmission.
BR-EL-035	Task Assignment Notification NOTIFICATION	System generates notification whenever an actionable workflow task is assigned to an actor (Substitute, Unit Head, Approver).
BR-EL-036	Role-Based Authorization ACCESS	All ELM capabilities and endpoints require explicit role-based authorization verification.
BR-EL-037	Admin Maintenance != Sanction INTEGRITY	Authorized administrative record maintenance/correction does not create a sanctioning decision.
BR-EL-038	Configurable SLA Thresholds CONFIG	Reminder and escalation hour thresholds must be configurable by Leave Administrators.

5. Complete Use Case Specifications (EL-UC-001 through EL-UC-021)

Use Case ID	Name & Actor	Preconditions & Trigger	Main Flow & Terminal Outcome
EL-UC-001	Apply for Leave **Employee**	Employee authenticated; sufficient balance; no overlap. *Trigger:* Employee submits form.	Validates category, dates, calendar, and substitute. Initial status: `PENDING_SUBSTITUTE` (if sub), `APPROVED` (if Director), or `PENDING_UNIT_HEAD`.
EL-UC-002	Respond to Substitute **Substitute**	Request in `PENDING_SUBSTITUTE` and actor is nominated substitute.	Substitute accepts (→ `PENDING_UNIT_HEAD` or `APPROVED`) or declines (→ `SUBSTITUTE_DECLINED`).
EL-UC-003	Review at Unit Head **Unit Head (HoD)**	Request in `PENDING_UNIT_HEAD` or `CANCEL_REQUESTED`.	• CL/RH: Final `APPROVE` (→ `APPROVED`) or `REJECT`. • SCL/EL/COL/VL: Records `RECOMMENDED` / `NOT_RECOMMENDED` (→ `PENDING_SANCTION`).
EL-UC-004	Verify & Route Estt. **DR (Establishment)**	Request in `PENDING_ESTABLISHMENT` (where configured).	Verifies service record and forwards to competent authority (→ `PENDING_SANCTION`).
EL-UC-005	Final Sanction Decision **Sanctioning Authority**	Request in `PENDING_SANCTION`.	Approver records final Approve (→ `APPROVED`, debits ledger) or Reject (→ `REJECTED`, frees earmarked balance).
EL-UC-007	Withdraw Pending Request **Employee**	Request in any pre-final pending status.	Submits withdrawal reason. Immediate transition to `WITHDRAWN`; halts remaining approvals; clears dates.
EL-UC-008	Request Cancellation **Employee**	Request is `APPROVED` and `current_date < start_date`.	Submits reason → `CANCEL_REQUESTED`. Follows approval hierarchy. If approved → `CANCELLED` and balance refunded.
EL-UC-009	Request Extension **Employee**	Request is `APPROVED`, category is EL/COL/VL, and leave is ongoing.	Enters extended end date → `EXT_REQUESTED`. Follows approval hierarchy. Incremental days posted on approval.
EL-UC-010	Submit Resumption **Employee**	Request is `APPROVED` for EL/COL/VL and employee returns.	Enters actual resumption date and remarks → `RESUMED`. Routes to Sanctioning Authority.
EL-UC-011	Review Resumption **Sanctioning Authority**	Request in `RESUMED`.	Confirms return date. Applies BR-EL-028 adjustments (early refund / late debit). Status → `RESUMPTION_REVIEWED`.
EL-UC-012	View Balances & History **Employee**	Authenticated employee.	Returns annual ledger breakdown (credited, availed, remaining balance) and timeline history of past requests.
EL-UC-013	Export/Print Application **Employee**	Accessible leave request.	Generates printable/exportable document with applicant metadata, station leave details, and approval trail.
EL-UC-014	Maintain Policy & Hierarchy **Leave Admin**	Leave Admin authenticated.	Configures category entitlements, carry-forward rules, and unit-head/higher-sanction routing chains.
EL-UC-015	Publish Holiday Calendar **Leave Admin**	Leave Admin authenticated.	Adds/edits closed holidays and Restricted Holidays (RH); publishes annual calendar used by BR-EL-003/031.
EL-UC-016	Sync Offline Leave **Leave Admin**	Authorized offline paper sanction reference exists.	Records offline leave. Validates balance, deducts ledger, marks directly `APPROVED` (`offline_entry=True`, no routing).
EL-UC-017	Maintain Leave Records **Leave Admin**	Discrepancy or audit correction required.	Adjusts ledger balances with mandatory audit rationale without creating a sanctioning decision (BR-EL-037).
EL-UC-018	Configure SLA Thresholds **Leave Admin**	Admin authenticated.	Updates reminder hours (e.g. 48h) and escalation hours (e.g. 96h) per workflow task type.
EL-UC-019	Replace Substitute **Employee**	Request in `SUBSTITUTE_DECLINED`.	Nominates a replacement eligible substitute. Request returns to `PENDING_SUBSTITUTE`.
EL-UC-020	Execute Year-End Processing **Event Triggered**	Annual boundary (December 31 midnight).	• Lapses unused CL, RH, and SCL. • Converts unused faculty VL to EL at **2:1 ratio**. • Carries forward EL and COL balances.
EL-UC-021	Process SLA Threshold Event **Event Triggered**	Periodic scheduler execution.	Dispatches reminder notification if pending at reminder threshold; executes escalation/re-routing at escalation threshold.

6. Workflows & State Transition Specifications

6.1 Composite Workflows (CW-EL-01 through CW-EL-04)

- **CW-EL-01 (New Leave Processing & Pending Withdrawal):** Complete end-to-end progression for new applications. Incorporates substitute consent gate/recovery loop, Unit Head final decision for CL/RH, higher-sanction routing for SCL/EL/COL/VL, Director self-sanction, and immediate withdrawal from every pending pre-final state.
- **CW-EL-02 (Approved Leave Cancellation):** Covers pre-start cancellation initiated by the employee. Progresses through the original approval hierarchy. Approved cancellation marks request `CANCELLED` and refunds balance; rejected cancellation retains `APPROVED`.
- **CW-EL-03 (Ongoing Leave Extension):** Governs requests for additional leave time during ongoing EL, COL, or VL. Passes through Unit Head recommendation and final sanction. Approval credits only incremental additional days.
- **CW-EL-04 (Resumption Review):** Manages post-leave return to duty. Employee submits actual resumption date. Sanctioning Authority confirms return, distinguishing normal return, early return (balance refund), and late return (extra deduction).

6.2 Complete State Transition Table (STT)

Current State	Action / Event	Conditions & Authority	Next State	Ledger & System Side Effects
`DRAFT`	Submit Application	Substitute required	`PENDING_SUBSTITUTE`	Audit log created; substitute notified (BR-EL-035).
`DRAFT`	Submit Application	No substitute; Director applicant	`APPROVED`	Auto-sanction (BR-EL-020); balance debited.
`DRAFT`	Submit Application	No substitute; Normal employee	`PENDING_UNIT_HEAD`	Audit log created; Unit Head notified.
`PENDING_SUBSTITUTE`	Substitute Accept	Normal employee	`PENDING_UNIT_HEAD`	Consent gate opens; routes to Unit Head.
`PENDING_SUBSTITUTE`	Substitute Accept	Director applicant	`APPROVED`	Director self-sanction completes; balance debited.
`PENDING_SUBSTITUTE`	Substitute Decline	Nominated substitute	`SUBSTITUTE_DECLINED`	Consent gate closed; applicant notified.
`SUBSTITUTE_DECLINED`	Replace Substitute	Applicant nominates valid peer	`PENDING_SUBSTITUTE`	New nominee assigned; recovery loop begins.
`PENDING_UNIT_HEAD`	Unit Head Approve	Category in (CL, RH)	`APPROVED`	Final sanction; balance deducted from ledger.
`PENDING_UNIT_HEAD`	Unit Head Reject	Category in (CL, RH)	`REJECTED`	Final rejection; application terminated.
`PENDING_UNIT_HEAD`	Unit Head Recommend	Category in (SCL, EL, COL, VL)	`PENDING_SANCTION`	Forwards to final competent authority.
`PENDING_UNIT_HEAD`	Unit Head Not Recommend	Category in (SCL, EL, COL, VL)	`PENDING_SANCTION`	Forwards to final authority with negative note.
`PENDING_SANCTION`	Sanction Approve	Final Sanctioning Authority	`APPROVED`	Final sanction; balance deducted from ledger.
`PENDING_SANCTION`	Sanction Reject	Final Sanctioning Authority	`REJECTED`	Final rejection; application terminated.
`Any Pending State`	Immediate Withdraw	Applicant before final approval	`WITHDRAWN`	Workflow terminates; tasks cleared (BR-EL-022).
`APPROVED`	Request Cancel	`current_date < start_date`	`CANCEL_REQUESTED`	Routes through original approval hierarchy.
`CANCEL_REQUESTED`	Approve Cancel	Competent Authority	`CANCELLED`	Days refunded to balance ledger (BR-EL-023).
`CANCEL_REQUESTED`	Reject Cancel	Competent Authority	`APPROVED`	Leave remains approved; no balance adjustment.
`APPROVED`	Request Extension	Ongoing EL/COL/VL	`EXT_REQUESTED`	Incremental days routed through hierarchy.
`APPROVED`	Submit Resumption	Return from EL/COL/VL	`RESUMED`	Routes to Sanctioning Authority for review.
`RESUMED`	Confirm Resumption	Sanctioning Authority	`RESUMPTION_REVIEWED`	Ledger adjusted per BR-EL-028 (normal/early/late).

7. Phase-2 Code Review & Comprehensive Unit Test Strategy

7.1 Audit Findings on Inherited Phase-1 Implementation

An execution of `pytest modules/leave/tests/` reveals that the inherited test suite fails during test collection due to fatal import errors:

```
ImportError: cannot import name 'create_leave_request' from 'modules.leave.services'
```

- **Obsolete Stubs:** `test_models.py` and `test_services.py` import outdated scaffold methods that do not exist in the refactored domain service layer.
- **Uninitialized Fixtures:** Test cases invoke `self.user` without instantiating test users in `setUp()`.
- **Architectural Boundary Violation:** Tests attempt to pass `user=self.user` into `LeaveRequest.objects.create()`, which violates the strict integer `user_id` isolation rule.
- **Missing Test Coverage:** Zero automated tests exist for day calculations, continuous counting, overlap exceptions, substitute consent gates, Unit Head recommendations, pre-start cancellations, resumption adjustments, or year-end conversions.

7.2 Production-Ready Unit & Integration Test Catalog

The following comprehensive test suites must be implemented under `modules/leave/tests/`:

Test Module	Target Functionality & Rules	Assertions & Verified Outcomes
test_calculator.py *Domain Calculator*	• Half-Day CL (BR-EL-010) • Continuous Counting (BR-EL-011) • Working-Day Counting (BR-EL-011)	• Half-day CL consumes exactly 0.5 days on a single date. • Requesting half-day on non-CL or across multiple dates raises `ValueError`. • Fri-Mon interval for EL/COL/VL returns 4.0 days (weekends count). • Fri-Mon interval for CL/SCL/RH returns 2.0 days (weekends excluded). • Closed institute holidays in `HolidayCalendar` are excluded for CL.
test_rules.py *Business Rules Validation*	• Balance Checks (BR-EL-002..009) • Faculty VL Scope (BR-EL-006) • Restricted Holidays (BR-EL-003) • Overlap & Half-Day Exception (BR-EL-012) • Substitute Conflicting Leave (BR-EL-014)	• Applying with insufficient balance raises `ValueError`. • Staff employee applying for VL raises eligibility error. • VL requested outside published `VacationPeriod` raises error. • RH date not in published calendar raises validation error. • Overlapping active leaves raises error; complementary half-day CL on the same date succeeds. • Nominating self or substitute with conflicting leave raises error.
test_state_machine.py *Pure Python Workflows*	• CW-EL-01 Initial Status • Substitute Accept/Decline Loops • Unit Head vs Final Sanction • Immediate Withdrawal (BR-EL-022)	• Initial status is `PENDING_SUBSTITUTE` if substitute nominated. • Director applicant auto-sanctions to `APPROVED`. • Substitute decline transitions to `SUBSTITUTE_DECLINED`; replacement returns to `PENDING_SUBSTITUTE`. • Unit Head `APPROVE` yields `APPROVED` for CL/RH, but `PENDING_SANCTION` for EL/COL/VL. • `can_withdraw()` returns True for all pending states and False for terminal states.
test_services.py *ORM & Transactions*	• Transactional Application Submission • Sanction Balance Deduction • Cancellation Balance Refund • Resumption Adjustments (BR-EL-028) • Year-End Processing (BR-EL-007)	• `apply_leave()` creates request and audit log row atomically. • Sanction approval decrements `balance` and increments `availed` in `leave_balances`. • Cancellation approval changes status to `CANCELLED` and refunds balance. • Early resumption refunds days; late resumption deducts extra days. • Year-end lapses CL/RH and converts unused VL to EL at 2:1 ratio.
test_api.py *Security & Permissions*	• IamSessionAuthentication • HasModuleGrant Protection • RBAC Role Restrictions • Approvals Inbox Queueing	• Unauthenticated request returns 401 Unauthorized. • Principal missing `leave_module` grant returns 403 Forbidden. • Employee cannot access Unit Head approval endpoint (403 Forbidden). • Approver with `leave_module.request.review` successfully reviews request.